Numerical based on unit-4 (Parsing and POS tagging using HMM)

Parsing context free grammar (CFG) using Top down or Bottom up approach

Constructing parse tree using given context free grammar rules

Question:

You are given the following CFG:

S → NP VP

NP → Pron | Det N

VP → V NP

Det → 'the'

N → 'dog' | 'cat'

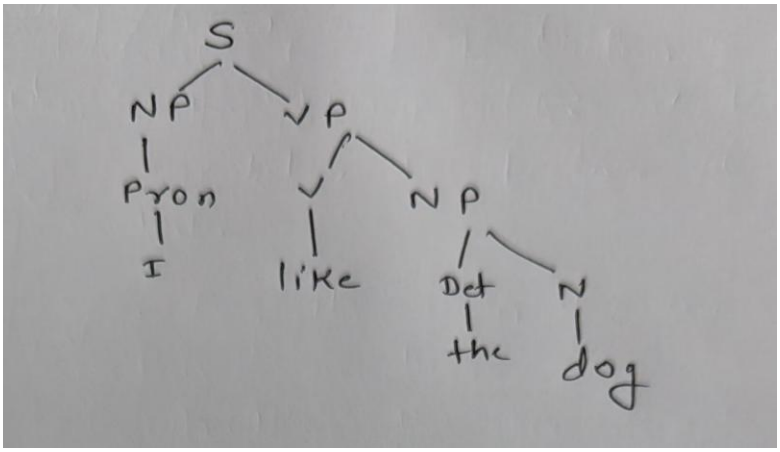
Pron → 'I'

V → 'like' | 'saw'

Draw the parse tree for the sentence:

“I like the dog.”

Solution :



Question:

Using the CFG below:

S → NP VP

NP → Det N

VP → V NP

Det → 'the'

N → 'boy' | 'girl'

V → 'met'

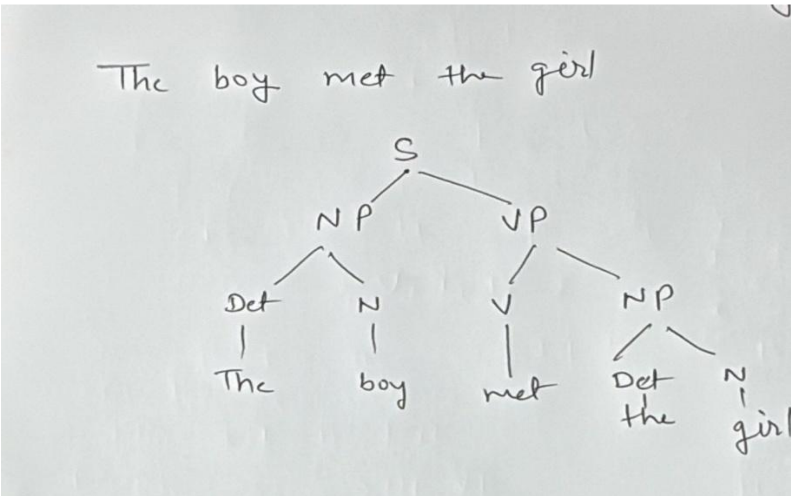
a) Draw the parse tree for the sentence:

“The boy met the girl.”

b) Explain why this sentence is unambiguous under the given grammar.

Solution:

Only one parse exists:



Explanation:

- **The grammar restricts NP → Det N and VP → V NP.**
- **There is no rule for attaching PP to either NP or VP, so no structural ambiguity is possible.**

Question: with ambiguous sentence

The sentence *“I saw the man with the telescope”* can be parsed in two different ways as shown in the trees below. Explain the syntactic ambiguity in this sentence using the given parse trees. What are the two possible meanings of the sentence according to the parses?

First interpretation/meaning- I saw the man. I was on the hill. I was using a telescope

CFG is given below:(Grammar where the PP attaches to the VP)

S → NP VP

NP → Pron | Det N

VP → V NP | VP PP

PP → P NP

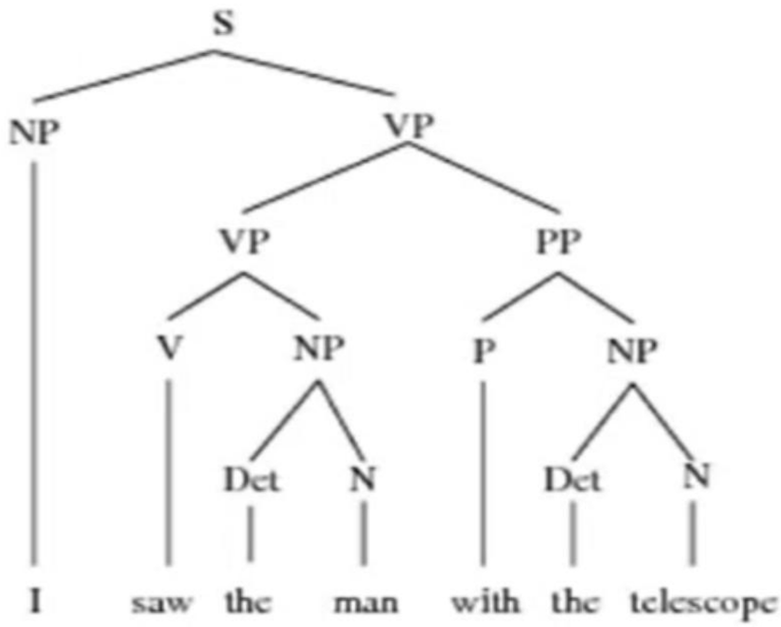
Pron → 'I'

V → 'saw'

Det → 'the'

N → 'man' | 'telescope'

P → 'with'



Explanation: VP → VP PP allows the PP with the telescope to attach to the VP (saw the man) producing the reading "I [saw the man] [with the telescope]" where the PP modifies the action.

For second interpretation/ meaning- I saw the man. I was on the hill. The hill had a telescope.

CFG is given below (Grammar where the PP attaches to the NP)

S → NP VP

NP → Pron | Det N | Det N PP

VP → V NP

PP → P NP

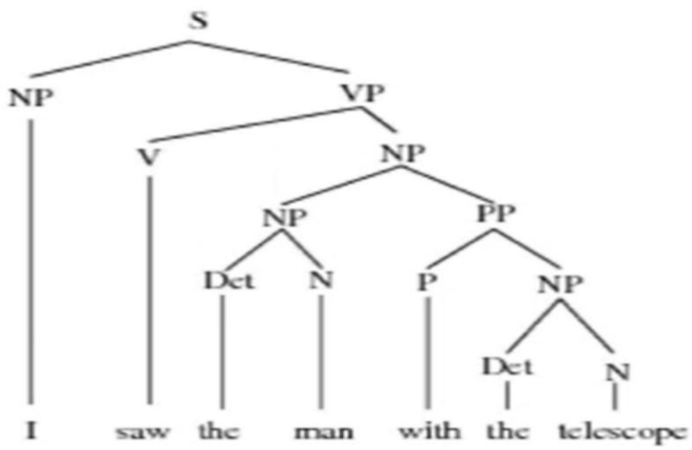
Pron → 'I'

V → 'saw'

Det → 'the'

N → 'man' | 'telescope'

P → 'with'



Explanation: NP → Det N PP allows the NP the man with the telescope so the PP is internal to the noun phrase (the man with the telescope), giving the reading "I saw [the man with the telescope]".

Question : Draw the parse tree(s) for "I met the girl in the park." using the given CFG.

S → NP VP

NP → Pron | Det N | Det N PP

VP → V NP | VP PP

PP → P NP

Pron → 'I'

V → 'met'

Det → 'the'

N → 'girl' | 'park'

P → 'in'

Solution: For the meaning : ***The meeting (met) occurred in the park.***

S

├─ NP

│ └─ Pron: I

└─ VP

├─ VP

| ├– V: met

| └─ NP

| ├– Det: the

| └─ N: girl

└─ PP

├– P: in

└─ NP

├– Det: the

└─ N: park

Parse tree for the meaning: The girl (whom I met) is the one in the park.

S

├– NP

| └─ Pron: I

└─ VP

├– V: met

└─ NP

├– Det: the

├– N: girl

└─ PP

├– P: in

└─ NP

├– Det: the

└─ N: park

Question based on HMM:

Forward Probability in HMM

You have an HMM with hidden states {Rainy, Sunny} and observations {walk, shop}.

Transition probabilities:

- P(Rainy → Rainy) = 0.7, P(Rainy → Sunny) = 0.3
- P(Sunny → Rainy) = 0.4, P(Sunny → Sunny) = 0.6

Emission probabilities:

- P(walk | Rainy) = 0.1, P(shop | Rainy) = 0.4
- P(walk | Sunny) = 0.6, P(shop | Sunny) = 0.3

Question:

Compute the **forward probability** that the observation sequence:

walk → **shop** is generated by this HMM, starting from Rainy with probability 1.

Solution:

Step 1: Initialization (t=1, "walk")

- α_1(Rainy) = 1 × P(walk|Rainy) = 1 × 0.1 = 0.1
- α_1(Sunny) = 0 (since start in Rainy only)

Step 2: Induction (t=2, "shop")

- α_2(Rainy) = [α_1(Rainy) × P(Rainy→Rainy)] × P(shop|Rainy)
  = 0.1 × 0.7 × 0.4 = 0.028
- α_2(Sunny) = [α_1(Rainy) × P(Rainy→Sunny)] × P(shop|Sunny)
  = 0.1 × 0.3 × 0.3 = 0.009

Step 3: Total probability = α_2(Rainy) + α_2(Sunny)

= 0.028 + 0.009 = **0.037**

Answer: Probability of observing *walk* → *shop* = **0.037**

Problem: Forward Algorithm in HMM

You are given a Hidden Markov Model with two hidden states:

Rainy (R)

Sunny (S)

Transition probabilities

P(R → R) = 0.7, P(R → S) = 0.3

P(S → R) = 0.4, P(S → S) = 0.6

Emission probabilities

P(walk | R) = 0.1, P(shop | R) = 0.4, P(clean | R) = 0.5

P(walk | S) = 0.6, P(shop | S) = 0.3, P(clean | S) = 0.1

Initial probabilities

P(R) = 0.6

P(S) = 0.4

Question:

Use the Forward Algorithm to compute the probability of the observation sequence:

O = {walk, shop, clean}

Step 1: Initialization (t = 1, observation = *walk*)

$$\alpha_1(R) = P(R) \times P(walk|R) = 0.6 \times 0.1 = 0.06$$

$$\alpha_1(S) = P(S) \times P(walk|S) = 0.4 \times 0.6 = 0.24$$

Step 2: Induction (t = 2, observation = *shop*)

$$\alpha_2(R) = [\alpha_1(R)P(R \to R) + \alpha_1(S)P(S \to R)] \times P(shop|R)$$

$$= [(0.06 \times 0.7) + (0.24 \times 0.4)] \times 0.4$$

$$= (0.042 + 0.096) \times 0.4 = 0.138 \times 0.4 = 0.0552$$

$$\alpha_2(S) = [\alpha_1(R)P(R \to S) + \alpha_1(S)P(S \to S)] \times P(shop|S)$$

$$= [(0.06 \times 0.3) + (0.24 \times 0.6)] \times 0.3$$

$$= (0.018 + 0.144) \times 0.3 = 0.162 \times 0.3 = 0.0486$$

Step 3: Induction (t = 3, observation = *clean*)

$$\alpha_3(R) = [\alpha_2(R)P(R \to R) + \alpha_2(S)P(S \to R)] \times P(clean|R)$$

$$= [(0.0552 \times 0.7) + (0.0486 \times 0.4)] \times 0.5$$

$$= (0.03864 + 0.01944) \times 0.5 = 0.05808 \times 0.5 = 0.02904$$

$$\alpha_3(S) = [\alpha_2(R)P(R \to S) + \alpha_2(S)P(S \to S)] \times P(clean|S)$$

$$= [(0.0552 \times 0.3) + (0.0486 \times 0.6)] \times 0.1$$

$$= (0.01656 + 0.02916) \times 0.1 = 0.04572 \times 0.1 = 0.004572$$

Step 4: Termination

The total probability of the observation sequence:

$$P(O) = \alpha_3(R) + \alpha_3(S) = 0.02904 + 0.004572 = 0.033612$$

Final Answer

The probability of observing the sequence **{walk, shop, clean}** is:

$$\boxed{0.0336\ (approx.)}$$

Named Entity Recognition with HMM

Consider the sentence:

“Paris is beautiful.”

You are given the HMM:

Transition probabilities (Tag → Tag)

- P(START → Location) = 0.6
- P(START → O) = 0.4
- P(Location → O) = 1.0
- P(O → O) = 0.8
- P(O → END) = 0.2

Emission probabilities (Word | Tag)

- P(“Paris” | Location) = 0.7
- P(“Paris” | O) = 0.3
- P(“is” | O) = 0.6
- P(“beautiful” | O) = 0.7

Question:

Use HMM to determine whether “Paris” should be tagged as a **Location** or not.

Solution:

Step 1: Initialization

- P(Location for “Paris”) = 0.6 × 0.7 = 0.42
- P(O for “Paris”) = 0.4 × 0.3 = 0.12

Step 2: Next word = “is” (O)

- From Location → O: 0.42 × 1.0 × 0.6 = 0.252
- From O → O: 0.12 × 0.8 × 0.6 = 0.0576

Step 3: Next word = “beautiful” (O)

- From O → O: 0.252 × 0.8 × 0.7 = 0.14112
- From O → O (second path): 0.0576 × 0.8 × 0.7 = 0.032256

Final best score = 0.14112 (path with “Paris” = Location).

Answer: “Paris” is most likely tagged as **Location**.

POS Tagging with HMM

Problem: POS Tagging with HMM

You are given the following HMM for **Part-of-Speech (POS) tagging**.

Tag set:

- **N** = Noun
- **V** = Verb

Initial probabilities:

- P(N) = 0.6
- P(V) = 0.4

Transition probabilities:

- P(N → N) = 0.3, P(N → V) = 0.7
- P(V → N) = 0.8, P(V → V) = 0.2

Emission probabilities:

- P("dog" | N) = 0.5, P("dog" | V) = 0.1
- P("barks" | N) = 0.1, P("barks" | V) = 0.6

Question:

For the sentence **"dog barks"**, compute the probability of **each possible tag sequence** using the HMM. Which sequence is more likely?

Solution

The sentence has 2 words:

- Word_1 = "dog"
- Word_2 = "barks"

Possible tag sequences = {N N, N V, V N, V V}.

1. Sequence = N N

$$P = P(N) \times P(dog|N) \times P(N \to N) \times P(barks|N)$$

$$= 0.6 \times 0.5 \times 0.3 \times 0.1 = 0.009$$

2. Sequence = N V

$$P = P(N) \times P(dog|N) \times P(N \to V) \times P(barks|V)$$

$$= 0.6 \times 0.5 \times 0.7 \times 0.6 = 0.126$$

3. Sequence = V N

$$P = P(V) \times P(dog|V) \times P(V \to N) \times P(barks|N)$$
$$= 0.4 \times 0.1 \times 0.8 \times 0.1 = 0.0032$$

4. Sequence = V V

$$P = P(V) \times P(dog|V) \times P(V \to V) \times P(barks|V)$$
$$= 0.4 \times 0.1 \times 0.2 \times 0.6 = 0.0048$$

Final comparison

- P(N N) = 0.009
- P(N V) = 0.126 (highest)
- P(V N) = 0.0032
- P(V V) = 0.0048

Answer

The most likely POS tag sequence for **“dog barks”** is:

$$\boxed{Noun\ \ Verb}$$

Interpretation: *“dog” = Noun, “barks” = Verb.*